Chapter 13
Gaining Support Within Universities for Collaboration

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13.1 Introduction

We have developed a village concept around which a main street and town square exist and function to build community living with neighborhoods and friendly neighbors. It is a social model of active living with life purpose, rather than the traditional institutional model of medical care (Schlegel: 2015 as cited in the University of Waterloo Daily Bulletin).

Imagine walking into a space where the quality of life and care for seniors has been thoroughly reimagined. This space is also a living laboratory and a place to

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R. Hall, A. Boccanfuso (eds.), *University-Industry Collaboration*, International Series in Operations Research & Management Science 369,
https://doi.org/10.1007/978-3-031-94913-5_13

receive hands-on training. Additionally, it is a facility where industry and academia work together across disciplines toward a shared goal—helping older adults thrive. For faculty members, this is an opportunity to see the real-world impact of their research in action. Research that is not only enhancing the lives of local community members but is also working toward a new paradigm of aging that can be used as a model worldwide. This is the Schlegel-UW Research Institute for Aging (RIA), one of the University of Waterloo's successful challenge-based partnerships.

This chapter explores incentivization to engage in university–industry (UI) research while providing practical solutions to challenges identified and reducing barriers to empower faculty to seize these opportunities. Fundamental and exploratory research is at the core of a research-intensive university; however, with ongoing changes and limitations to traditional research funding pathways, additional sources to maintain an individual academic's research program have emerged as a key support, such as more applied research with industry partners. Some faculties across campus readily see the value in partnering with industry while others may be more reluctant in developing relationships with for-profit organizations. This can be due to perceived concerns around intellectual property and publication rights, misalignment on objectives, confidentiality, reputational impact, and conflict of interest to name a few examples.

The field of UI collaboration is complicated. While some literature focuses narrowly on "contract research," the broader field understands the deep impact that these collaborations bring to both universities and industry, as a "bi-directional exchange of information" (Ankrah & Al-Tabbaa, 2015, p. 388). While this chapter provides general discussion on UI collaborations, the authors all hold positions in research administration at the University of Waterloo. We span academic administrators, those holding faculty appointments alongside administrative appointments, as well as specialists in corporate partnerships and government relations. Ultimately, while the principles we articulate are broadly applicable across the post-secondary sector, they ultimately stem from our own experiences at the University of Waterloo. Some context on our institution is thus important.

The University of Waterloo is a relatively young institution, founded in 1959. From the outset, and not without controversy, the University of Waterloo was predicated on strong links with industry. Indeed, the university's first president Joseph Gerald "Gerry" Hagey came not from the traditional halls of academia but rather from the business community; he was National Advertising Director of B. F. Goodrich. While at the tire manufacturer, as noted by the UW Archives, Hagey had become interested in the idea of students working in their respective industries while studying believing that it would provide experience and revenue for the students, revenue for the college, and assistance for the company (Hagey: 1957 as cited by Goel, 2022). This idea forged the core of Hagey's vision of education, resulting in the University of Waterloo today having North America's largest cooperative education program. This vision of close industrial collaboration within undergraduate programs extended to faculty research, including an inventor-owned IP policy that has become a key factor in faculty member recruitment as well as local economic development. It also meant that "industry" is in some ways part of the ethos of the University of Waterloo. As noted below, however, this does not insulate the institution from many of the broader challenges, including disciplinary cultures and perceived conflicts of interest.

13.2 Opportunities, Challenges, and Solutions

13.2.1 Opportunities

> *Industry-University partnerships are widely seen as important elements of innovative economies. We really have to get industries and universities working together.* —National Academy of Sciences President Marcia McNutt

> *By fostering collaboration between industry and academia, we ensure Canada remains at the forefront of technological advancement and economic growth.* —François-Philippe Champagne, Minister of Innovation, Science and Industry of Canada

Having government and industry-based incentives for UI collaborations is an essential cornerstone to support an innovation environment of collaboration (Sjöo & Hellström, 2019; Tseng et al., 2018). Currently, in Canada, several key STEM grant funding programs are predicated on dollar-for-dollar matching funds from industry partners (e.g., the Natural Science and Engineering Research Council (NSERC) Alliance Advantage program and the Mitacs Accelerate Program). In some cases, by stacking different programs, researchers can get as much as 3:1 leverage on industry funding, providing a powerful incentive for academics to interest industry in their research activities/capabilities. The other incentive these types of collaborative funding programs have is training High Quality Personnel (HQP), in the form of undergraduate, graduate, and postdoctoral fellow funding to work alongside these industry partners; in turn, providing opportunities for industry participants to build close relationships with the students and thus building their talent pipelines. Additionally, these programs provide faculty members unique training opportunities for their mentees to apply their theoretical and technical skills in real-life settings, which can also be a unique feature for student recruitment.

By contrast, grant programs that do not have an industry partnership requirement are more competitive. To provide a sense of scale, over the past five years at the University of Waterloo, one of the Faculties' research funding total value from industry and industry-leveraged programs through granting councils was $134 million Canadian Dollars (CAD) out of a total $389 million CAD. The amount of funding from NSERC over that period was $115.5 million CAD of which $47.7 million CAD (41%) was from the industry-matched Alliance Advantage Program. The extent to which industry and leveraged funding constitute a major component of total research funding (30–40%, depending on the baseline used) shows the impact of this structural incentivization for researchers to seek industry funding.

A particular opportunity for the University of Waterloo is our emphasis on student experiential learning. The university boasts one of the largest undergraduate co-op programs in the world, annually linking 26,000 undergraduate co-op students with more than 8000 domestic and international employers. However, this type of work-integrated learning opportunity with industry is practically and logistically more difficult for research-focused students in graduate programs. It is important to note that Canada's economy is based predominantly on small and medium enterprises as well as Canadian branch plants and offices of foreign multinational corporations, with fewer global head offices, making it a challenge to identify appropriate paid research-focused internship experiences for student trainees.

This training challenge has been addressed to a significant extent by the Mitacs Accelerate Program. Accelerate supports industry placements for graduate students or postdoctoral fellows that are linked to a university-based research project managed by an academic principal investigator (Mitacs pays roughly 50% of project costs tied to a match from industry). The actual residency time of the student will vary from project to project, but there is an expectation of at least some interaction on the company site for the trainees participating in the program with the hopes that the trainees will gain an appreciation for research challenges from an industry perspective. Regardless of that, students can reap the benefits of applying their academic research skills through working on an applied project aimed at technology development or problem-solving in an industrial context. Due to the success of this program, Mitacs has grown to support over 20,000 work-integrated learning placements annually. In Waterloo's case, investing in an expert co-funded with Mitacs to leverage this support has resulted in significant uptake of the Accelerate program, further bolstering the university's experiential learning advantage.

The benefits of participation in these government-supported UI collaborative research partnership programs are many (Ankrah & Al-Tabbaa, 2015, Sjöo & Hellström, 2019, Tseng et al., 2018; Fig. 13.1).

Overall, graduate internship programs have proven to provide a significant incentivization for faculty members to partner with industry on research projects and vice versa.

Finally, the University of Waterloo has a track record of encouraging nontraditional partnerships. In part, this is from the founding ethos of the institution. As Gerald Hagey, the University of Waterloo's first president, argued, "I cannot perceive a time when the universities will not be challenged by new requirements from our society. Equally, I cannot foresee a time when the University of Waterloo will be so hidebound by tradition that it cannot adjust itself to providing education to meet these needs" (Hagey: 1957 as cited by Goel, 2022)

In 2022, the University of Waterloo developed the Global Futures Framework that seeks to accelerate collaborative opportunities through an interdisciplinary approach to the university's educational programs, research, and innovation activities. The five interconnected futures (societal, health, sustainable, technological, and economic) aligned to the University of Waterloo's academic and research strengths, help focus and coordinate the institution's work across disciplines and organizational boundaries. Each future overlaps with the others and it is in these intersections where Waterloo's strength for identifying new opportunities, solutions, and leading change will be best leveraged for maximum impact.

Another focal point of the UI interface was the recent creation of Waterloo Ventures. By joining the University of Waterloo entities that focus on technology

Academic supervisor	Student	Industry
• Research budgets from industry maximized with support of leveraged funding programs • Practical application of knowledge and insight into new challenge areas for technology development or research • New contacts and professional network development • Opportunity to provide graduate students experience outside of academic labs	• Work-integrated learning and experience • Training that may be applicable to non-academic career pathways • Development of a professional network • Insight into industry priorities and challenges	• Financial leverage on R&D spending • Talent pipeline development • Support from an expert academic (at no additional cost) • Tax incentives for R&D investment • Rapid solutions to critical problems • Onsite support for research • Building of essential collaboration for future support

Fig. 13.1 Stakeholder benefits to UI partnership

transfer (Waterloo Commercialization Office) and the Velocity incubator, and through engaging thought leadership and social impact (GreenHouse and Grebel Peace Incubator(s), Waterloo is solidifying its role as a leader within Canada. These efforts help students, researchers, staff, and founders mobilize their ideas toward real-world impact, often through commercialization efforts. Waterloo Ventures creates awareness of industry needs as it relates to R&D and commercialization from within an academic setting.

13.2.2 Challenges and Solutions: How to Motivate Faculty

A new focus for universities beyond the creation of knowledge is to expand networks and transfer research into commercialized products and services (Sjöo & Hellström, 2019). To incentivize university faculty members to participate in UI collaborations, it is valuable to investigate barriers to involvement in a partnership. Bruneel et al. (2010) identifies two categories: (i) differences in UI orientation and (ii) transaction-related barriers.

13.3 UI Orientation

The culture of the university is a key factor in encouraging these types of partnerships (Sjöo & Hellström, 2019; Tseng et al., 2018). As was previously discussed, the University of Waterloo has created a unique industry-minded innovation ecosystem through the cooperative education program, our incubators, and our creator-owned intellectual property policy. In addition to encouraging collaborative opportunities from its inception, having a large faculty of engineering, also minded toward industry relevancy, helps drive the spirit of industry collaboration, which has shown to be self-reinforcing (Sjöo & Hellström, 2019).

The University of Waterloo demonstrates its commitment to research and innovation partnerships in several ways. For example, we have an active membership in the National Cybersecurity Consortium. This is a national network supporting the Canadian cybersecurity sector, encouraging collaboration among private industry, not-for-profit organizations, provincial, territorial, and municipal governments alongside academia. As a Founding Member, the University of Waterloo is supporting projects that nurture talent development, innovation, technology adoption, and commercialization.

The second category of the barriers identified by Bruneel et al. (2010) is focused on the transactional support for these collaborations. High level of support and resourcing creates an environment that can help streamline these challenges (Nsanzumuhire & Groot, 2020). This can include "active enablers" such as Technology Transfer Offices, Industry Liaison Offices, Incubators, and Collaborative Research Centres. These offices help facilitate partnership development by minimizing "cognitive distance and increasing social and physical proximity" (Nsanzumuhire & Groot, 2020). The administrators in these offices have critical skills and expertise in navigating UI relationships and navigating conflict (Sjöo & Hellström, 2019, Nsanzumuhire & Groot, 2020, Bruneel et al., 2010). As a Canadian Institution, without the implications of the Bayh-Dole Act (except in projects under US authority), the University of Waterloo creates a passive enabling structure by incentivizing innovation-minded researchers to think of the applications of their IP beyond the lab bench (Sjöo & Hellström, 2019; Bruneel et al., 2010).

A central Office of Research support system provides high-level support and guidance for research collaboration as it relates to university-level policies and strategy development that may take into account multiple campus stakeholders. Central-level administrators act as ambassadors, representing the university as a whole, as well as serving as a natural point of contact for major industry partners. This provides administration the ability to find industry sector-based opportunities that may be interdisciplinary, and even inter-faculty, as a central office is able to observe all the research activity on campus. The university-level office can also act as a concierge to more faculty-specific opportunities. Faculty-level supports can provide specialized support attuned to their own faculty-member disciplines, where opportunities could be niche in focus. There is a deeper knowledge of what each individual faculty member's interests are and can match specific industry needs with potential academic collaborators quickly. An approach where both functions work collaboratively provides a seamless structure for both faculty members and industry to gain support based on their needs quickly and efficiently.

Implementing collaborations with the private sector in non-STEM fields has particular challenges. Understanding social science will often be critically important to assessing and managing societal impact and acceptance of new (e.g., AI)—or old (e.g., vaccines)—technologies. However, because individual companies do not have a proprietary interest in the results of this kind of research (notwithstanding that it may determine the survival of entire industry sectors, e.g., GMOs), it is often difficult to get private-sector funding for this sort of research. Moreover, the traditional siloed university structure leads to significant challenges in multidisciplinary research that requires bridging disparate academic cultures and practices between STEM and non-STEM researchers.

Amongst the six individual faculty units at the University of Waterloo (Arts, Environment, Health, Engineering, Mathematics, Science), each has a culture that may lend itself to, or away from, research partnerships. Generally speaking, it is culturally expected that engineering faculty will have applied research activities that often lend itself quite easily to industry partnerships (Bruneel et al., 2010). There are many supports at the university to help encourage and support these collaborations such as staff hired specifically to support external partnerships and technology transfer officers dedicated to the faculty. Industry partnerships have an expectation of funding support from the external partner for the particular projects.

As a large percentage of our faculty on campus is in engineering, our central support unit of the Office of Research provides additional support and coordination due to the volume of activity through our industry liaison and technology transfer offices. On the other hand, the Faculty of Arts research partnerships are more often with not-for-profit or government organizations. Due to lower volume of collaborations in these fields, there may be a lower level of support to assist faculty in cultivating new partnerships as attention is directed toward more active engagements. It can be difficult to maintain an appropriate balance of support across faculties as the university has a goal in increasing knowledge-sharing with resource-limited pressures, to "remain at the leading edge in all subject areas" (Ankrah & Al-Tabbaa, 2015). This is an area where the administrative offices can provide additional resources to support capacity-building for partnership development, through hosting networking events with external partners. We also offer training workshops to demystify partnerships and offer one-on-one support and guidance on building partnerships for their specific needs. What is clear to those facilitating partnerships is that each faculty must be supported based on their individualized needs and goals, requiring significant consultation before developing partnership plans.

One of the University of Waterloo's marquis partnerships is with a national communications company. This multifaceted relationship was initiated through alignment in deep technological understanding of 5G technology. Over time, the relationship grew to include research collaborations with additional industry partners acting as technology adopters allowing for a broader range of researchers at the university to participate. These additional opportunities require strong account management from the central Office of Research to help facilitate networking events, workshops, student activities like hackathons, and discussions on broad topic areas before focusing on specific research outcomes.

Some researchers are reluctant to develop partnerships and may have concerns and misconceptions around academic freedom and fear of being constrained in the dissemination of results when collaborating with industry (Sjöo & Hellström, 2019;

Ankrah & Al-Tabbaa, 2015). Additionally, researchers may feel overwhelmed by the contracting process, specifically when negotiating a research contract that includes complex intellectual property and confidentiality terms. There has not been any evidence of correlation between these concerns and the ability to partner (Sjöo & Hellström, 2019). Having administrators well-versed in these aspects of academic contracting to support the culture of partnership between academics and industry can be confidence-boosting to the faculty involved and also takes the burden off of researchers for handling these dimensions of the partnership arrangement. Through the active support of industry liaison and partnership grant managers within the University of Waterloo Office of Research, there have been some key capacity-building opportunities for researchers in this space. One of these supports is an award-winning workshop for early-career researchers entitled "Planning Your Research Trajectory," a training and information session directed at new (and new to the university) faculty members (CAUBO, 2020). Although it includes several portfolios within the Office of Research, it provides tangible information to help combat preconceived concerns about partnership. It provides these researchers an opportunity to learn about the support and guidance available by the administrators within the Office of Research to support their academic integrity. A strong contracting team can help minimize these concerns by discerning academics from commercial confidential information, and to retain rights to any IP for academic purposes.

Location can influence the decision to engage in UI collaborations (Sjöo & Hellström, 2019; Bruneel et al., 2010). Larger centers provide more access to a variety of partners and potentially a more mature ecosystem. Factors in place-based innovation ecosystems are inherently unique but always include a combination of stakeholders such as universities, government, established industry, startup companies, and investors:

> Within one hour's drive is the engine that drives the economy in Ontario and the economy in Canada.—Simon Drexler, General Manager, Automation Product

The University of Waterloo is embedded in the Toronto-Waterloo Corridor, which has several strategic advantages for collaboration, including access to 15,000 tech companies, over 315,000 tech workers, 16 universities and colleges and world-leading medical, engineering, and artificial intelligence experts (Waterloo EDC, n.d.). A mature innovation ecosystem is the major factor in the ability to have successful partnerships (Sjöo & Hellström, 2019). Not only is there a large number of potential collaborators, but having access to these companies within a short commute allows for collaborative research relationships to build through regular, meaningful interactions with entire academic and industrial teams, as well as providing opportunities to engage beyond one specific project, building deeper relationships, and minimizing difficulties such as managing time zones where resolving issues with local partners can be much more straightforward.

Although a part of a bustling industrial corridor, the unique geographical position of the University of Waterloo places it within a semirural setting. This juxtaposition creates an understanding of rural needs and opportunities to apply technological solutions to assist, such as agricultural applications of robotics, drone technologies, and environmental modelling. The Region of Waterloo has ample space for industry to invest by creating automotive manufacturing plants and aerospace-focused R&D offices. These investments create opportunities to further enhance the local innovation ecosystem. The University of Waterloo has built a strong relationship with the Region of Waterloo airport to help facilitate further UI collaboration in this space.

Faculty may struggle to find the time and resources needed to develop relationships and consider it a burden added to their already busy academic schedules. By strategically taking advantage of the university's location, a key function of the Office of Research partnership team is to develop and execute connector events on campus. By working with individual faculties to determine needs, areas of interest, and potential external participants, we help ensure interest from faculty in attending and reduce barriers by having these events on campus and in person. After the COVID-19 pandemic, it is evident through post-event surveys of faculty and external participants that face-to-face interaction is of high value when building a foundation of trust and respect for partnerships to develop. Additional opportunities arise from co-locating industry personnel on campus or even in specific labs of collaborating academic researchers, to increase interaction and exposure as an opportunity to build more meaningful relationships such as research development clusters, research parks, consortia, use of specialized equipment and services, as well as less research-intensive activities such as participation in job fairs, advisory boards, and sponsorships.

One of the ways the University of Waterloo has initiated co-location is through the development of the National Research Council of Canada (NRC)—University of Waterloo Collaboration Centre. The NRC supports business innovation and academic knowledge by working with partners in industry, government, and academia. This hub was developed for students, government, and faculty members to connect and collaborate in the areas of artificial intelligence, Internet of Things, and cybersecurity. Building HQP skills and experience, and creating an environment where diverse perspectives can be shared and learned from are instrumental in developing technologies with the goals of commercialization or industry use in mind.

The values and merits of potential industrial partners need to be considered by the university when deciding to enter (or not enter) into a given relationship. First are of course adherence to laws and strictures and government sanctions. Yet beyond legal compliance come moral and ethical considerations, as well as the values of the university. Should environmental, social, and governance factors enter the consideration? Recent moves toward climate change divestment within university investment strategies show that universities understand the moral weight of the relationships that they enter.

The University of Waterloo administration has taken a leadership role in supporting researchers navigating the complexities of today's environment. Initial stages included several opportunities for education and sharing of information on the topic with researchers and other academicians. The Office of Research hosted a conference entitled "Research Security in Today's Geo-Political Era" that served to share concerns of stakeholders, learning more about the threats to research from foreign actors, and hearing about research security plans from government funding agencies as well as government officials. Further supplementing this workshop, a team has been recruited to support the University of Waterloo's Safeguarding Research initiative. This team within the Office of Research provides ongoing support to researchers and their international research affiliations, engagements, and collaborations considering transparency, predictability, academic freedom, and norms of open science (University of Waterloo, n.d.). With this team as an additional resource for faculty interested in partnership, faculty who may have had concerns related to geopolitical tension can have confidence in the support of the university when developing these partnership opportunities.

From a partnership development perspective, industry has rigid goals and expectations based on budgets and timelines. They are typically looking to solve a specific problem. For an academic researcher, there is significant specificity in the area of research specialization, perhaps having a limited view of a project slightly outside of their scope (Nsanzumuhire & Groot, 2020). Bridging this divide requires flexibility on both sides. Partnership managers can act as this bridge to help translate value. By identifying the particular set of skills of a researcher, the partnership manager can counsel a researcher and help identify areas within a company where they could be applied, and vice versa. By understanding an industry partner's needs, and uncovering the areas of novelty, acting as a conduit for that information back to a faculty member can help simplify these initial discussions. It is often difficult to pitch a particular project to industry without having a prior relationship due to some of the aforementioned constraints; however, through developing broader university-level partnerships, and through individual partnership manager professional networks, there are often opportunities to share academic opportunities to those open to learning more. Conversely, for academic researchers to be interested in an industry project, partnership managers can help discern the value add to a researcher by investigating the opportunity before presenting it internally at the university. The partnership manager can be seen as a trusted source for opportunities by stringent vetting the potential project to determine if it is a good fit for the university and the researchers who may participate. Considerations typically include the company history, ability to fund a research project, interest in true academic research (not exclusively development), and bandwidth of researchers in the area of interest).

One obstacle to UI partnerships is, of course, our model of research assessments within the academy. Faculty members have their work primarily structured around the needs of tenure and promotion, which is a decision that is made in conjunction by university administrations with the disciplinary "guilds" that make up the university. While formal criteria will differ between different institutions, at research universities the research output of a faculty member will be a main factor (alongside teaching, and to a lesser extent, their role in service of the university, discipline, and society) in them being granted tenure. Each discipline sets the "bar" differently, but this could range from a scholarly monograph in many arts disciplines to high-impact journal articles and conference proceedings in others. The pressures are such that many researchers can be forgiven for thinking that any research activity *not* in service of these outputs is time wasted.

There has been a recent trend across the research ecosystem to note that this type of incentivization leads to several deleterious effects. Most notably, it undervalues certain kinds of work: time-consuming community-based work, the creation of research datasets, leadership in the development of regulatory standards, and—germane to this chapter—commercialization activities. Patents, technology transfer, and industrial collaboration all have a broader impact on society and economies beyond just the peer-reviewed publications. Commercialization-based sabbaticals for faculty can provide the support and time needed for faculty to explore the commercialization of their research by joining corporate entities for a period of time. It provides the researcher a deeper understanding of the complexities of integrating their technology into the current ecosystem, and when they return to campus, can bring back insights, ideas, and opportunities learned during their sabbatical. This deepens ties between the two organizations and could even lead to co-funded appointments, where world-class researchers have a foot in industry and another on-campus training the next generation of innovators.

Recent declarations, such as the Declaration on Research Assessment (DORA), seek to underscore the importance of impact consideration, rather than the rote and uncritical use of bibliometric indicators. The journal name itself is not that important, or even the existence of a journal itself, in the DORA view of how to evaluate a researcher, but rather what impact the research and activity has had. A more holistic view of research assessment also helps to address other issues across the research ecosystem, including Equity, Diversity, and Inclusion (EDI): developing partnerships is time consuming, and we know that women, racialized groups and those with disabilities face especially pressing pulls on their time, both external and internal to the university.

13.4 Case Study: The Schlegel-UW Research Institute for Aging

The Schlegel-UW Research Institute for Aging (RIA) is one representative example of an impactful partnership at the University of Waterloo. Initiated in 2005, this large-scale interdisciplinary partnership grew out of extensive discussions between the University of Waterloo and Schlegel Villages—the largest, for-profit retirement living network in Ontario. The overarching goal of the RIA is to foster healthy aging by "changing the way we age in Canada." This partnership is of high societal significance because the quality of life for the elderly population (over six million in Canada and growing) is well recognized as a priority for the Canadian healthcare system. Guided by this premise, the RIA seeks to generate and mobilize new knowledge and technology that address the physical, social, emotional, and mental dimensions of aging to improve care practices, healthcare services, and educational training for the senior living sector.

The University of Waterloo and Schlegel Villages are working together to rally around the societal grand challenge of aging. This challenge is far too complex with too many needs and considerations to be tackled by one entity entirely. By building this nontraditional, multifaceted approach, a coalition of stakeholders can work together to drive the transformational research to help tackle aging with perspectives of different stakeholders with research and technology application with immediate impact and directly with the end users (e.g., patient or clinician). Through this interface, a testbed environment has been created, allowing researchers and clinicians to receive immediate feedback, refine ideas, and review research results (Ankrah & Al-Tabbaa, 2015). The testbed creates a collaborative environment to find solutions that truly work and to share them for the benefit of older adults everywhere (RIA, 2023). Having a specific mission focused around aging can inspire researchers to be directly involved in research that is impactful globally.

Sustained, holistic engagement of stakeholders is the key reason for the successful initiation and realization of the RIA partnership. Both the university and the external partner have invested substantially to bring the initiative to fruition. The Schlegel family, who founded the Schlegel Villages, has been deeply engaged with the University of Waterloo from the start to co-design the blueprint and development roadmap of the RIA together with university administrators and stakeholders. They have stepwise invested over $50M over the years to support the RIA building construction at the university's North Campus, establish 12 sponsored chair professorships, and support research activities related to aging. Members of the Schlegel family are also frequent participants in RIA research activities and events, and they remain involved as board members to help steer the ongoing growth of the RIA. The Schlegel Villages and its residents are also firsthand knowledge users of fruitful research outcomes by RIA researchers.

Generous funding support provides a strong foundation for the partnership. By faculty leveraging the relationship with the RIA, additional support can be provided both in-kind and through additional funding contracts, and as a partner on other funding agency grants. Our researchers' networks can also expand by connecting with partners the RIA can bring to the table. This expanded relationship capability provides additional opportunities for our faculty members and their trainees to enhance their academic efforts. By having a hub of innovation for aging, including other academic partners and a network of external end users, researchers can see the value in participating in this type of initiative to further their own research agendas.

The training of graduate students and clinical care professionals is a core mission of the RIA. Under the supervision of RIA researchers who are full-time faculty members at the University of Waterloo, graduate students participate in a variety of interdisciplinary research projects in aging. The RIA also partners with Conestoga College, based in the Region of Waterloo for nursing training.

A Statistics Canada Economic and Social Report from 2022 indicated that almost two thirds of PhD graduates in Canada worked in occupations outside of academia. Having an external partner supporting graduate student trainees on-site offers students the chance to engage in research outside of a typical academic research environment. Faculty members therefore have this opportunity to provide this type of experience to their students and can use this opportunity not only as a recruitment tool for top-notch talent, but also set their current trainees up for success upon graduation. The University of Waterloo culture of industry partnership can be seen through work-integrated learning extending well beyond the typical co-op model.

Since becoming operational in 2014, the RIA has brought together a multidisciplinary team of researchers from the Faculty of Health, Faculty of Engineering and Faculty of Arts to collaboratively tackle major research problems in aging. The unique physical environment of the RIA allows clinicians, scientists, and engineers to work alongside each other to address a broad range of aging research topics, including cardiovascular health, dementia, food and nutrition, geriatric medicine, mental health, mobility, spirituality, and care team development. The RIA also serves well as an innovation hub for devising and deploying novel technologies for senior living and routine diagnostics.

Due to the unique nature of this partnership, having representation of researchers from three of the University of Waterloo's six faculties, awareness of the success of the partnership has grown across the campus. The university is championing a significant high-value partnership where researchers from various fields can see themselves and their research reflected in real-world outcomes. The success stories have been shared to highlight successes across disciplines and combat distinct faculty cultures where partnership may be seen in a less positive light. The faculty members working within the RIA can be seen as role models and mentors amongst the peers within their own faculties, helping break down partnership barriers further for academicians.

One key advantage provided by the RIA to researchers is its strong knowledge mobilization network through the Schlegel Villages. For example, upon admission to the retirement homes of Schlegel Villages, new residents have an opportunity to participate in the Schlegel Functional Fitness Assessment scheme. This assessment focuses on physical and motor capabilities and helps guide the recommendations for physical activities provided by the Registered Kinesiologist who works in every Schlegel Village. New technologies developed at the RIA, such as new measures of cerebrovascular health biomarkers, have been added to this assessment scheme to accelerate the research impact of new diagnostic technologies to the aging population.

The RIA Model for Innovation (Fig. 13.2) includes four phases, identified in their 2022–2023 Impact Report:

1. Knowledge generation
2. Incubation
   - Discovering and developing innovations
3. Acceleration
   - Expanding and evaluating innovations in new settings
4. Mobilization
   - Sharing knowledge to the benefits of older adults everywhere

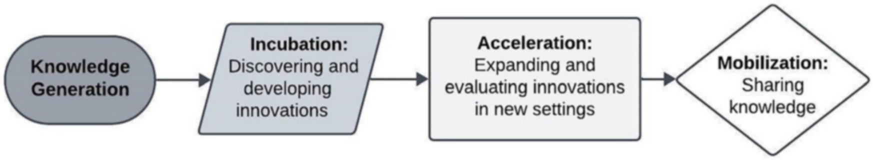

Fig. 13.2 RIA model for innovation

This unique relationship reaps the benefits of co-location effectively. There are immediate opportunities for knowledge creation, scaling opportunities and mobilization activity, supported by immediate feedback on outcomes. Technology feedback and adoption are near immediate as researchers and end users can walk down the hall to see the research developments in action.

13.5 Takeaways and Future Strategies

The RIA is an example of how rethinking UI partnerships can provide an entry point for researchers less inclined to participate in these types of collaborations. Interdisciplinary research in a co-location setting breaks down barriers, improves credibility amongst stakeholders, and provides a dynamic environment that promotes a bidirectional flow of ideas that benefits all parties. It can also offer shared infrastructure costs, potential for commercialization, student training, talent attraction, startup creation, and enhanced external funding opportunities (UIDP, 2017).

The University of Waterloo has cultivated a culture of partnership with industry and provided transactional support through human resources, as well as effective policies and procedures, keeping external stakeholders in mind. President Hagey's vision has blossomed under careful consideration of the impact of the knowledge created within our Ivory Towers. By creating inclusive spaces where everyone is part of the solution, the University of Waterloo's Global Futures vision is well poised to tackle the big challenges within our society. In a rapidly evolving world, institutions must look at the impact and application of technology, not just the development of technology. The world continues to face multiple, compounding major crises that cut across society, health, the environment, technology, and the economy. These complex challenges are felt locally and globally. They are pressing opportunities for us to work with others toward new approaches to teaching and learning and to the creation of new ideas and collaborative solutions. Incentivizing researchers to think beyond the laboratory and by extension, the gates of the university, to support partnerships with immediate, real-world applicability benefits us all.

The University of Waterloo has a flourishing cooperative education program, an innovation-driving intellectual property policy, and support to ensure industry investments in research can be leveraged to add additional value to the projects being developed. In order to motivate faculty and encourage participation in partnered research, barriers such as building connections and finding value propositions and reducing friction such as having strong administrative support will reorient an academic culture to one of knowledge mobilization, translation, and impact with industry.

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